Moller scattering 4

From the first section we have

$$\sum_{abcd} \mathcal{M}_1 \mathcal{M}_1^* = \frac{e^4}{t^2} f_{11}$$
$$\sum_{abcd} \mathcal{M}_1 \mathcal{M}_2^* = \frac{e^4}{tu} f_{12}$$
$$\sum_{abcd} \mathcal{M}_2 \mathcal{M}_2^* = \frac{e^4}{u^2} f_{22}$$

where

$$f_{11} = 8s^2 + 8u^2 - 64m^2 s - 64m^2 u + 192m^4$$
$$f_{12} = -8s^2 + 64m^2 s - 96m^4$$
$$f_{22} = 8s^2 + 8t^2 - 64m^2 s - 64m^2 t + 192m^4$$

Densities can also be formed as

$$f_{11} = 32(p_1 \cdot p_2)^2 + 32(p_1 \cdot p_4)^2 - 64m^2(p_1 \cdot p_2) + 64m^2(p_1 \cdot p_4)$$
$$f_{12} = -32(p_1 \cdot p_2)^2 + 64m^2(p_1 \cdot p_2)$$
$$f_{22} = 32(p_1 \cdot p_2)^2 + 32(p_1 \cdot p_3)^2 - 64m^2(p_1 \cdot p_2) + 64m^2(p_1 \cdot p_3)$$

For $m = 0$

$$32(p_1 \cdot p_2)^2 = 8s^2$$
$$32(p_1 \cdot p_3)^2 = 8t^2$$
$$32(p_1 \cdot p_4)^2 = 8u^2$$

Eigenmath script